

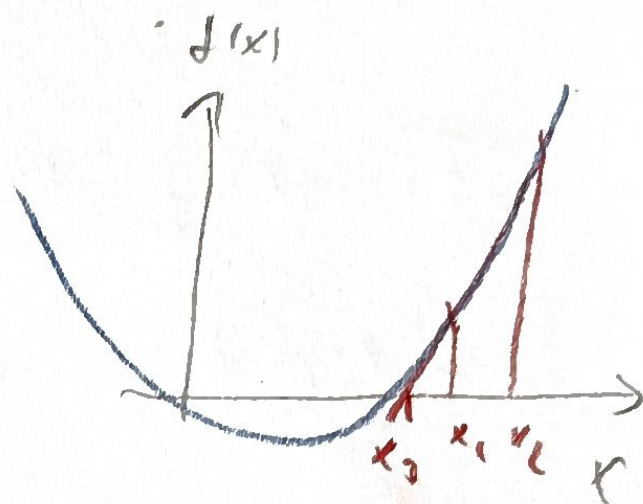
Introduction to numerical Methods - 9 -

1

False position method (regula falsi)

Predict the position of the root by
linear approximation

$$\frac{f(x_3) - f(x_2)}{x_3 - x_2} \approx \frac{f(x_1) - f(x_2)}{x_1 - x_2}$$



Set $f(x_3) = 0 \Rightarrow$

$$x_3 = x_2 - f(x_2) \frac{x_1 - x_2}{f(x_1) - f(x_2)}$$

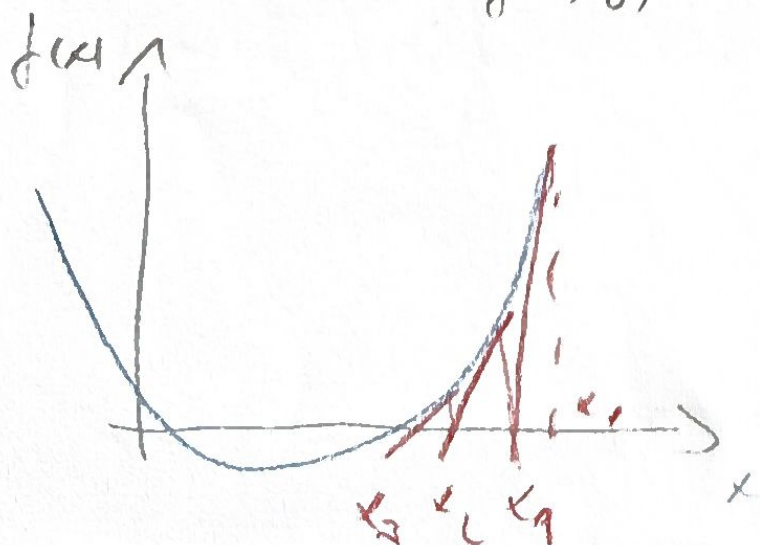
- Much faster than bisection
- Works with starting interval without a root

Newton's method: single point method,

use the local gradient

$$0 = f(x_0) + f'(x_0)(x_1 - x_0)$$

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$



- generally faster than regula falsi
- requires derivative

12

- If you know how to find minima then finding minima of $f^2(x)$ can give you the roots

Finding the minima

Gradient method:

Let $f(x)$ be a differentiable, and $\delta > 0$ then

$$x_{n+1} = x_n - \delta f'(x_n)$$

if x_n is not an extremal point then

$f(x_{n+1}) \leq f(x_n)$ if δ is sufficiently small

Conjugate gradient method

idea: close to the minima everything is a parabola: $f(x) \approx \beta(x-a)^2$
minimum a .

Two points x_1, x_2 , value $f(x_1), f(x_2)$
derivative $f'(x_1), f'(x_2)$

The gradient method:

$$x_2 = x_1 - \delta f'(x_1) = x_1 - 2\delta\beta(x_1 - a)$$

We want $x_2 = a \Rightarrow 2\delta\beta = 1 \Rightarrow \delta = \frac{1}{2\beta}$

So if we can make $\delta = \frac{1}{2\beta}$ the gradient method will go to the minimum

$$f'(x_1) = 2\beta(x_1 - a) \quad f'(x_2) = 2\beta(x_2 - a)$$

$$f'(x_2) - f'(x_1) = 2\beta x_2 - 2\beta a - 2\beta x_1 + 2\beta a = 2\beta(x_2 - x_1)$$

$$\delta = \frac{1}{2\beta} = \frac{x_2 - x_1}{f'(x_2) - f'(x_1)}$$

Conjugate method is a dimension

$$x_{n+1} = x_n - \beta_n f'(x_n)$$

$$\beta_n = \frac{(x_n - x_{n-1})^T (f'(x_n) - f'(x_{n-1}))}{\|f'(x_n) - f'(x_{n-1})\|^2}$$

this set is also a parabola



Problems

14

- local minima
- = stochastic processes
- etc.

Simulated annealing

- stochastic method
- able to break from local minimum

Idea: - allow unfavourable moves

- slowly cool down the system:
at large temperatures, the system does not care about the energy landscape, but as cooled gradually goes to valleys but has time to escape local minima



- ergodic moves? may reach all possible states

- new:

- $f(x_{\text{new}}) \leq f(x_{\text{old}})$: keep new pos
- $f(x_{\text{new}}) > f(x_{\text{old}})$: keep new pos with prob. $\exp(- (f(x_{\text{new}}) - f(x_{\text{old}})) / T)$

T temp and goes to 0 for $t \rightarrow \infty$